Excel Capstone Project - E-Commerce Analysis

PROJECT OBJECTIVE:

- **Data Cleaning:** Remove duplicate records and fill in missing values using appropriate formulas within the retail sales dataset.
- **Data Formatting & Standardization:** Standardize text data by removing extra spaces, spelling errors, and formatting inconsistencies using functions like PROPER, TRIM, and CLEAN.
- **Business Calculations:** Accurately calculate the total sales amount (TOTAL AMOUNT) based on quantity, unit price, and discount.
- **Data Modelling:** Establish a Star Schema structure by relating the Customer Dimension, Product Dimension, Store Dimension, and Sales Fact tables for efficient analysis.

DATASET DESCRIPTION:

This dataset consists of 4 main tables (dimensions and facts) representing the overall operations of a retail business:

1. **Customer Dim:** Contains details about customers such as their name, age, gender, location (city, state, country), and loyalty level.
2. **Product Dim:** Contains details about products sold, including product name, category, sub-category, brand, cost, and stock availability.
3. **Store Dim:** Contains details about store names, regions, cities, and store types (Flagship, Outlet, Online).
4. **Sales Fact:** Contains actual sales transaction records, including order dates, quantities, unit prices, discounts, and total amounts.

COLUMN DESCRIPTION:

The details of key columns across all tables are listed in the table below:

Column Name	Description
Sales ID	Unique identification code for each sales transaction.
Order Date	The date when the products were ordered or purchased.
Customer_ID	Unique ID used to identify a customer.
Product_ID	Unique ID used to identify a product.
Store ID	Unique ID used to identify the store where the sale took place.
Quantity	The number of units of a product purchased by the customer.
Unit Price	The selling price of a single unit of a product.
Discount	The discount percentage applied to that specific transaction.
TOTAL AMOUNT	The final calculated total sales amount after deducting the discount.
Name	Full name of the customer.
Age	Age of the customer.
Gender	Gender of the customer.
City	City where the customer resides.
State	State where the customer resides.
Country	Country where the customer resides.
Loyalty_Level	Customer's membership/tier status (Platinum, Gold, Silver, Bronze, UNKNOWN).
Product Name	Name of the product.
Category	Primary category of the product (Electronics, Clothing, Beauty, etc.).
Sub_Categ	Sub-category of the product (Computer, Outerwear, Skincare, etc.).
Brand	Name of the brand that manufactured the product.
Cost	Cost or procurement price of the product.
Stock	Number of items currently available in inventory/stock.
Store Name	Name of the store.
Region	Geographic region where the store is located (East, West, North, South).
Store Type	Type of the store (Flagship, Outlet, Online).

DATA CLEANING & TRANSFORMATION:

USED FORMULA:

- **REMOVE DUPLICATE DATA:**
  Data--->Data Tool--->Remove Duplicates

- **CORRECT FORMATTING:**
  (CUSTOMER DIM) : PROPER(TRIM(CLEAN(B2) ROPER(TRIM(CLEAN(E2)
  PROPER (TRIM(CLEAN(F2)
  PROPER(TRIM(CLEAN(H2)
  (PRODUCT DIM) : PROPER(TRIM(CLEAN(B2) ROPER(TRIM(CLEAN(C2)
  PROPER(TRIM(CLEAN(D2) PROPER(TRIM(CLEAN(E2)
  (STORE DIM) : PROPER(TRIM(CLEAN(B2) PROPER(TRIM(CLEAN(D2)

- **LOYALTY_LEVEL COLUMN HANDLE MISSING DATA:**
  IF(ISBLANK(H2),"UNKNOWN", H2)

- **COST AND STOCK COLUMN HANDLE MISSING DATA:**
  COST : IF(ISBLANK(F2), AVERAGE (F2:F103), F2),
  STOCK: IF(ISBLANK(G2), AVERAGE (G2:G103), G2)

- **QUANTITY, UNIT PRICE AND DISCOUNT COLUMNS HANDLE MISSING DATA with AVERAGE:**
  QUANTITY : IF(ISBLANK(F2), AVERAGE (F2:F2001), F2)
  UNIT PRICE : IF(ISBLANK(G2), AVERAGE (G2:G2001), G2)
  DISCOUNT : IF(ISBLANK(H2), AVERAGE (H2:H2001), H2)

SCREENSHOTS:

DESCRIPTIVE STATSTICS

DESCRIPTIVE STATISTICS			
Quantity		**Unit_Price**	
Mean	*2.4765*	*Mean*	*513.9831146*
Standard Error	*0.024890433*	*Standard Error*	*6.307795632*
Median	*2*	*Median*	*522.43*
Mode	*2*	*Mode*	*871.34*
Standard Deviatio	*1.113133992*	*Standard Deviati*	*282.0931964*
Sample Variance	*1.239067284*	*Sample Variance*	*79576.57147*
Kurtosis	*-1.347725858*	*Kurtosis*	*-1.189707573*
Skewness	*0.029733624*	*Skewness*	*-0.036751141*
Range	*3*	*Range*	*979.8*
Minimum	*1*	*Minimum*	*20.09*
Maximum	*4*	*Maximum*	*999.89*
Sum	*4953*	*Sum*	*1027966.229*
Count	*2000*	*Count*	*2000*

1. Quantity:
Mean (Average): 2.48
Median & Mode: 2
Range: 1 to 4 (Range: 3)
Total Sum: 4,953

2. Unit Price:
Mean (Average): 513.98
Median: 522.43
Mode: 871.34
Range: 20.09 to 999.89 (Range: 979.8)
Total Sum: 1,027,966.23
Overall Takeaway:
Across 2,000 recorded transactions, the average purchase quantity is 2.48 units (ranging fro Chart Area ts) at an average unit price of 513.98, A total value of 1,027,966.23.

PIVOT TABLE & CHARTS

Row Labels	Sum of Total_Amount
Beauty	322582
Clothing	377543
Electronics	505315
Home	422651
Sports	542010
Grand Total	**2170101**

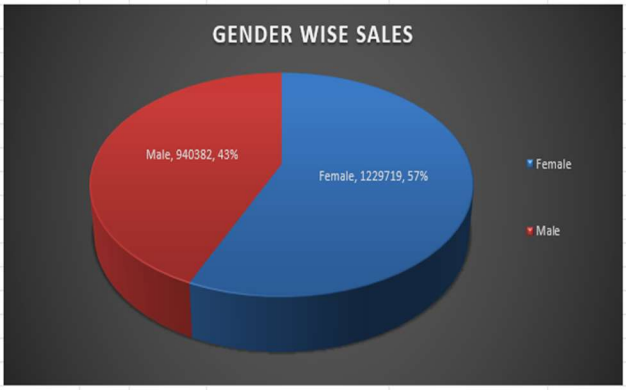


Row Labels	Average of Total_Amount
Aguirre-Keller	93048
Alexander Group	118814
Bean, Jones and I	158904
Benson-Elliott	115363
Black-Horn	98976
Frederick-Wright	111705
Gardner, Humphr	114593
Gray Ltd	80510
Guerrero-Schmitt	105437
Hill, Wagner and	109792
Holloway Ltd	120934
Johnson-Wright	123860
Mann-Austin	94226
Martinez, Adkins	99629
Martinez-Mullins	107987
Mcdonald Inc	106248
Moore, Padilla ar	108429
Rivera-Warner	85139
Shelton PLC	99679
Thomas, Bridges	116829
Grand Total	**2170101**

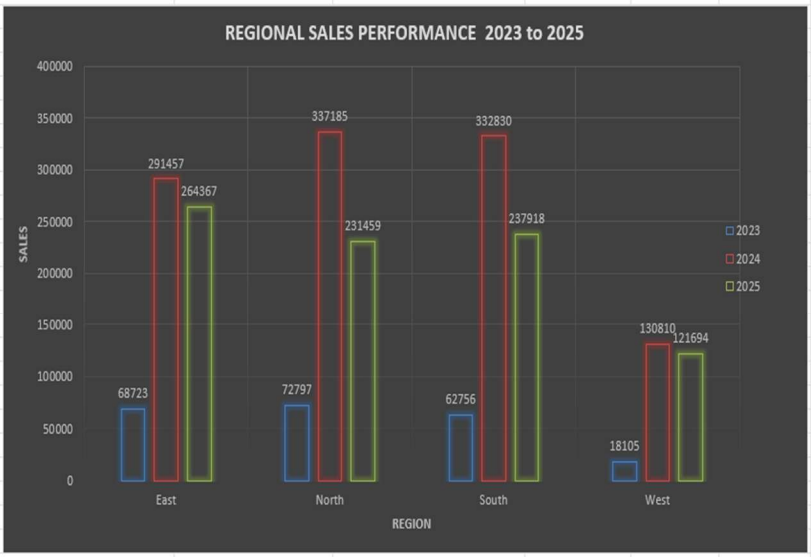


Grand Total	2170101

Row Labels	Sum of Total_Amount
Female	1229719
Male	940382
Grand Total	**2170101**



Sum of Total_Am	Column Labels			
	⊞2023	⊞2024	⊞2025	Grand Total
Row Labels				
East	68723	291457	264367	624548
North	72797	337185	231459	641440
South	62756	332830	237918	633503
West	18105	130810	121694	270609
Grand Total	**222381**	**1092282**	**855438**	**2170101**



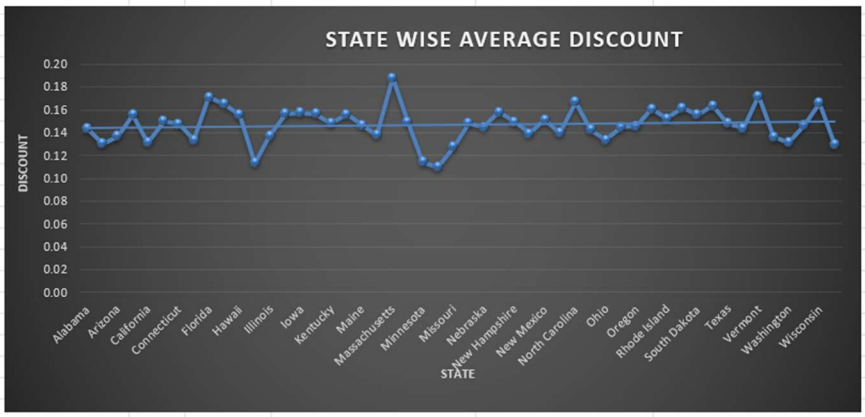
Row Labels	Sum of Quantity	Sum of Total_Amount
Flagship	1927	845543
Online	2090	914963
Outlet	936	409594
Grand Total	**4953**	**2170101**



Row Labels	Sum of Quantity
Bronze	1209
Gold	1189
Platinum	1485
Silver	1034
UNKNOWN	36
Grand Total	**4953**



Row Labels	Average of Discount
Alabama	0.14
Alaska	0.13
Arizona	0.14
Arkansas	0.16
California	0.13
Colorado	0.15
Connecticut	0.15
Delaware	0.13
Florida	0.17
Georgia	0.17
Hawaii	0.16
Idaho	0.11
Illinois	0.14
Indiana	0.16
Iowa	0.16
Kansas	0.16
Kentucky	0.15
Louisiana	0.16
Maine	0.15
Maryland	0.14
Massachusetts	0.19
Michigan	0.15
Minnesota	0.11
Mississippi	0.11
Missouri	0.13
Montana	0.15
Nebraska	0.15
Nevada	0.16
New Hampshire	0.15
New Jersey	0.14
New Mexico	0.15
New York	0.14
North Carolina	0.17
North Dakota	0.14
Ohio	0.13
Oklahoma	0.14
Oregon	0.15
Pennsylvania	0.16
Rhode Island	0.15
South Carolina	0.16



Xlookup using and Creating Table

Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	TOTAL AMOUNT	Customer Name	Product_Name	Store Name
SALE0	07-07-2024	CUST1409	PROD97	STORE3	1	218	0.09	198.42	Sarah Smith	Cricket Bat	Johnson-Wright
SALE1	25-12-2023	CUST1467	PROD74	STORE17	3	104	0.27	228.79	Michael Foley	Bluetooth Car Audio System	Holloway Ltd
SALE2	Chart Area	CUST1056	PROD49	STORE8	4	914	0.25	2742.15	Christopher Sutton	Men's Roll-On Perfume	Bean, Jones and Park
SALE3	29-11-2023	CUST1346	PROD76	STORE8	1	922	0.16	774.07	John Jordan	Wireless Mouse	Bean, Jones and Park
SALE4	06-01-2024	CUST1493	PROD59	STORE5	3	745	0.18	1833.44	Amanda Powell	Men's Hair Styling Gel	Gray Ltd
SALE5	26-07-2024	CUST1116	PROD15	STORE7	3	147	0.19	357.23	Anthony Phillips	4K Home Projector	Martinez-Mullins
SALE6	12-07-2025	CUST1016	PROD56	STORE4	2	785	0.15	1335.03	John Scott	Premium Blanket Throw	Alexander Group
SALE7	18-11-2023	CUST1152	PROD70	STORE8	4	342	0.23	1054.56	Patrick Roberts	Kitchen Storage Rack	Bean, Jones and Park
SALE8	04-05-2025	CUST1187	PROD99	STORE5	1	448	0.07	416.87	Dr. Suzanne Bullock	Stainless Steel Cookware Se	Gray Ltd
SALE9	31-07-2025	CUST1464	PROD33	STORE0	2	873	0.23	1344.68	David Pierce	Modern Wall Clock	Rivera-Warner
SALE10	13-02-2024	CUST1065	PROD24	STORE14	1	279	0.07	259.13	Michael Thomas	Foldable Camping Chair	Mcdonald Inc
SALE11	24-05-2025	CUST1320	PROD39	STORE1	4	287	0.17	952.57	Jessica Gonzalez	Portable Power Bank	Martinez, Adkins and Johnson
SALE12	24-08-2024	CUST1237	PROD88	STORE4	4	645	0.01	2552.54	Wesley Short	Sports Wrist Sweatbands	Alexander Group
SALE13	06-08-2025	CUST1443	PROD34	STORE16	2	317	0.30	443.66	Adam Castro	Gaming Mechanical Keyboa	Frederick-Wright
SALE14	21-06-2025	CUST1348	PROD24	STORE15	4	403	0.29	1145.85	Sherry Cabrera	Foldable Camping Chair	Shelton PLC
SALE15	10-12-2024	CUST1435	PROD36	STORE9	1	871	0.04	836.49	Lindsay Beasley	Women's Summer Dress	Moore, Padilla and Butler
SALE16	13-09-2025	CUST1276	PROD65	STORE19	2	452	0.25	677.97	Arthur Warren	Adjustable Dumbbells 20kg	Mann-Austin
SALE17	11-12-2024	CUST1079	PROD27	STORE4	2	503	0.23	774.14	Molly Gardner	Decorative Floor Vase	Alexander Group
SALE18	13-08-2024	CUST1094	PROD39	STORE4	2	559	0.25	838.85	David Patterson	Portable Power Bank	Alexander Group
SALE19	18-03-2024	CUST1239	PROD60	STORE13	4	514	0.04	1973.61	Maria Hill	Luxury Perfume Spray	Gardner, Humphrey and Contreras
SALE20	05-07-2025	CUST1144	PROD72	STORE17	3	986	0.27	2159.43	Justin Cross	Professional Jump Rope	Holloway Ltd
SALE21	04-06-2024	CUST1396	PROD43	STORE18	1	590	0.22	459.86	Alexander Byrd	USB External Hard Drive	Guerrero-Schmitt
SALE22	12-04-2024	CUST1476	PROD69	STORE10	3	979	0.03	2850.11	Elizabeth Foster	Men's Sports Shorts	Hill, Wagner and Ballard
SALE23	23-06-2024	CUST1368	PROD12	STORE8	1	719	0.08	661.74	Miranda King	Hair Strengthening Shampo	Bean, Jones and Park
SALE24	08-09-2025	CUST1186	PROD67	STORE6	3	78	0.07	216.50	Jennifer Pruitt	Smart LED Light Strip	Benson-Elliott
SALE25	17-09-2024	CUST1381	PROD5	STORE3	4	853	0.14	2933.32	Erika Padilla	Men's Leather Belt	Johnson-Wright
SALE26	03-01-2024	CUST1213	PROD98	STORE8	4	156	0.15	532.28	Stuart Johnson	Portable Bluetooth Speaker	Bean, Jones and Park
SALE27	22-02-2025	CUST1200	PROD77	STORE8	3	815	0.17	2030.57	Anna Perez	WiFi Mesh Router	Bean, Jones and Park
SALE28	23-09-2025	CUST1148	PROD13	STORE2	2	74	0.11	132.09	Theresa Campbell	USB-C Fast Charger	Black-Horn
SALE29	11-01-2024	CUST1348	PROD99	STORE3	1	452	0.15	383.92	Sherry Cabrera	Stainless Steel Cookware Se	Johnson-Wright
SALE30	18-10-2023	CUST1028	PROD5	STORE19	1	429	0.13	373.56	Joshua Ryan	Men's Leather Belt	Mann-Austin
SALE31	28-10-2024	CUST1412	PROD53	STORE12	2	149	0.03	288.98	Shane Brown	Men's Track Pants	Thomas, Bridges and Harris
SALE32	01-11-2024	CUST1398	PROD75	STORE8	4	314	0.08	1154.89	Jennifer Johnson	Women's Ankle Boots	Bean, Jones and Park
SALE33	24-11-2024	CUST1453	PROD12	STORE6	2	717	0.15	1219.43	April Phillips	Hair Strengthening Shampo	Benson-Elliott
SALE34	16-10-2023	CUST1344	PROD79	STORE1	1	873	0.18	716.04	Evelyn Brown	Decorative Lamp Shade	Martinez, Adkins and Johnson
SALE35	09-08-2025	CUST1218	PROD16	STORE3	4	109	0.19	354.33	Julia Jackson	Beard Grooming Kit	Johnson-Wright
SALE36	09-12-2024	CUST1086	PROD5	STORE7	3	560	0.16	1409.99	Tammy Miller Md	Men's Leather Belt	Martinez-Mullins

Xlookup Used to Create a Table

Using Formula for Customer Name Column
xlookup(C2,Cutomer_Dim!A:A,Customer_Dim!B:B)

Using Formula for Product Name Column
xlookup(D2,Product_Dim!A:A,Product_Dim!B:B)

Using Formula for Product Name Column
xlookup(E2,Store_Dim!A:A,Store_Dim!B:B)

Vlookup using and Creating Table

Sales_ID	Order_Dat	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	TOTAL AMOUNT	CUSTOMER NAME	PRODUCT NAME	STORE NAME
SALE0	07-07-2024	CUST1409	PROD97	STORE3	1	218	0.09	198.42	Sarah Smith	Cricket Bat	Johnson-Wright
SALE1	25-12-2023	CUST1467	PROD74	STORE17	3	104	0.27	228.79	Michael Foley	Bluetooth Car Audio System	Holloway Ltd
SALE2	03-06-2024	CUST1056	PROD49	STORE8	4	914	0.25	2742.15	Christopher Sutton	Men's Roll-On Perfume	Bean, Jones and Park
SALE3	29-11-2023	CUST1346	PROD76	STORE8	1	922	0.16	774.07	John Jordan	Wireless Mouse	Bean, Jones and Park
SALE4	06-01-2024	CUST1493	PROD59	STORE5	3	745	0.18	1833.44	Amanda Powell	Men's Hair Styling Gel	Gray Ltd
SALE5	26-07-2024	CUST1116	PROD15	STORE7	3	147	0.19	357.23	Anthony Phillips	4K Home Projector	Martinez-Mullins
SALE6	12-07-2025	CUST1016	PROD56	STORE4	2	785	0.15	1335.03	John Scott	Premium Blanket Throw	Alexander Group
SALE7	18-11-2023	CUST1152	PROD70	STORE8	4	342	0.23	1054.56	Patrick Roberts	Kitchen Storage Rack	Bean, Jones and Park
SALE8	04-05-2025	CUST1187	PROD99	STORE5	1	448	0.07	416.87	Dr. Suzanne Bullock	Stainless Steel Cookware Set	Gray Ltd
SALE9	31-07-2025	CUST1464	PROD33	STORE0	2	873	0.23	1344.68	David Pierce	Modern Wall Clock	Rivera-Warner
SALE10	13-02-2024	CUST1065	PROD24	STORE14	1	279	0.07	259.13	Michael Thomas	Foldable Camping Chair	Mcdonald Inc
SALE11	24-05-2025	CUST1320	PROD39	STORE1	4	287	0.17	952.57	Jessica Gonzalez	Portable Power Bank	Martinez, Adkins and Johnson
SALE12	24-08-2024	CUST1237	PROD88	STORE4	4	645	0.01	2552.54	Wesley Short	Sports Wrist Sweatbands	Alexander Group
SALE13	06-08-2025	CUST1443	PROD34	STORE16	2	317	0.30	443.66	Adam Castro	Gaming Mechanical Keyboard	Frederick-Wright
SALE14	21-06-2025	CUST1348	PROD24	STORE15	4	403	0.29	1145.85	Sherry Cabrera	Foldable Camping Chair	Shelton PLC
SALE15	10-12-2024	CUST1435	PROD36	STORE9	1	871	0.04	836.49	Lindsay Beasley	Women's Summer Dress	Moore, Padilla and Butler
SALE16	13-09-2025	CUST1276	PROD65	STORE19	2	452	0.25	677.97	Arthur Warren	Adjustable Dumbbells 20kg	Mann-Austin
SALE17	11-12-2024	CUST1079	PROD27	STORE4	2	503	0.23	774.14	Molly Gardner	Decorative Floor Vase	Alexander Group
SALE18	13-08-2024	CUST1094	PROD39	STORE4	2	559	0.25	838.85	David Patterson	Portable Power Bank	Alexander Group
SALE19	18-03-2024	CUST1239	PROD60	STORE13	4	514	0.04	1973.61	Maria Hill	Luxury Perfume Spray	Gardner, Humphrey and Contreras
SALE20	05-07-2025	CUST1144	PROD72	STORE17	3	986	0.27	2159.43	Justin Cross	Professional Jump Rope	Holloway Ltd
SALE21	04-06-2024	CUST1396	PROD43	STORE18	1	590	0.22	459.86	Alexander Byrd	USB External Hard Drive	Guerrero-Schmitt
SALE22	12-04-2024	CUST1476	PROD69	STORE10	3	979	0.03	2850.11	Elizabeth Foster	Men's Sports Shorts	Hill, Wagner and Ballard
SALE23	23-06-2024	CUST1368	PROD12	STORE8	1	719	0.08	661.74	Miranda King	Hair Strengthening Shampoo	Bean, Jones and Park
SALE24	08-09-2025	CUST1186	PROD67	STORE6	3	78	0.07	216.50	Jennifer Pruitt	Smart LED Light Strip	Benson-Elliott
SALE25	17-09-2024	CUST1381	PROD5	STORE3	4	853	0.14	2933.32	Erika Padilla	Men's Leather Belt	Johnson-Wright
SALE26	03-01-2024	CUST1213	PROD98	STORE8	4	156	0.15	532.28	Stuart Johnson	Portable Bluetooth Speaker	Bean, Jones and Park
SALE27	22-02-2025	CUST1200	PROD77	STORE8	3	815	0.17	2030.57	Anna Perez	WiFi Mesh Router	Bean, Jones and Park
SALE28	23-09-2025	CUST1148	PROD13	STORE2	2	74	0.11	132.09	Theresa Campbell	USB-C Fast Charger	Black-Horn
SALE29	11-01-2024	CUST1348	PROD99	STORE3	1	452	0.15	383.92	Sherry Cabrera	Stainless Steel Cookware Set	Johnson-Wright
SALE30	18-10-2023	CUST1028	PROD5	STORE19	1	429	0.13	373.56	Joshua Ryan	Men's Leather Belt	Mann-Austin
SALE31	28-10-2024	CUST1412	PROD53	STORE12	2	149	0.03	288.98	Shane Brown	Men's Track Pants	Thomas, Bridges and Harris
SALE32	01-11-2024	CUST1398	PROD75	STORE8	4	314	0.08	1154.89	Jennifer Johnson	Women's Ankle Boots	Bean, Jones and Park
SALE33	24-11-2024	CUST1453	PROD12	STORE6	2	717	0.15	1219.43	April Phillips	Hair Strengthening Shampoo	Benson-Elliott
SALE34	16-10-2023	CUST1344	PROD79	STORE1	1	873	0.18	716.04	Evelyn Brown	Decorative Lamp Shade	Martinez, Adkins and Johnson
SALE35	09-08-2025	CUST1218	PROD16	STORE3	4	109	0.19	354.33	Julia Jackson	Beard Grooming Kit	Johnson-Wright
SALE36	09-12-2024	CUST1086	PROD5	STORE7	3	560	0.16	1409.99	Tammy Miller Md	Men's Leather Belt	Martinez-Mullins

Chart Area

DESCRIPTIVE STATISTICS | PIVOT TABLE | DASHBOARD | xlookup | vlookup | EXPL OF X AND V LOOKUP CREATION

Vlookup Used to Create a Table

Chart Area

Using Formula for Customer Name Column
Vlookup(C2,Cutomer_Dim!A:B,2,False)

Using Formula for Product Name Column
Vlookup(D2,Product_Dim!A:B,2,False)

Using Formula for Product Name Column
Vlookup(E2,Store_Dim!A:B,2,False)

CUSTOMER DIM

Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level
CUST1000	Sheena Bonilla	55	Male	Donaldton	North Carolina	USA	UNKNOWN
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum
CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold
CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
CUST1011	Brian Lopez	28	Female	East Krista	New York	USA	Gold
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum
CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA	Bronze
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA	UNKNOWN
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA	Platinum
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA	Silver
CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA	Gold
CUST1022	Paul Fox	33	Female	Michaelton	Hawaii	USA	Silver
CUST1023	Jonathan Steele	52	Female	Molinabury	Oregon	USA	Silver
CUST1024	Heather Collins	40	Male	Rogerschester	Idaho	USA	Platinum
CUST1025	Diana Taylor	59	Female	Vasqueztown	California	USA	Bronze
CUST1026	Amber Carlson	18	Female	East Michellemouth	Missouri	USA	Gold
CUST1027	Leslie Johnson	31	Male	Mollyshire	Utah	USA	UNKNOWN
CUST1028	Joshua Ryan	27	Female	New David	New Jersey	USA	Platinum
CUST1029	David Arnold	54	Female	East Jennifer	California	USA	Silver
CUST1030	Jessica Acevedo	64	Female	Amybury	Pennsylvania	USA	Bronze

Customer_Dim | Product_Dim | Store_Dim | Sales_Fact | DESCRIPTIVE STATISTICS | PIVOT TABLE | DASHBOARD | xloo

Customer DIM Find, Remove Duplicates, Handle Missing, Inconsistent Data, Correct Formatting errors, and Validate Data

REMOVE DUPLICATE DATA: Data-->Data Tool-->Remove Duplicates

LOYALTY_LEVEL COLUMN HANDLE MISSING DATA:
IF(ISBLANK(H2),"UNKNOWN",H2)

CORRECT FORMATTING: PROPER(TRIM(CLEAN(B2)
PROPER(TRIM(CLEAN(E2) PROPER(TRIM(CLEAN(F2)
PROPER(TRIM(CLEAN(H2)

PRODUCT DIM

Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock
PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	112.42	416
PROD1	Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	342.56	181
PROD2	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc	254.62	352
PROD3	Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	364.67	452
PROD4	Women's Yoga Pants	Clothing	Bottoms	Gonzales-Davis	306.46	487
PROD5	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzale	261.00	59
PROD6	Herbal Face Moisturizer	Beauty	Skincare	Davidson PLC	312.80	350
PROD7	LED Table Lamp	Home	Lighting	Harrell LLC	218.86	170
PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	439.07	142
PROD9	Adjustable Dumbbell Set	Sports	Fitness	Hill PLC	495.72	323
PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	298.86	308
PROD11	HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weaver	449.40	271
PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	115.82	460
PROD13	USB-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242.23	453
PROD14	Decorative Wall Mirror	Home	Decor	Luna, Anderson and Bowma	48.01	439
PROD15	4K Home Projector	Electronics	Appliances	Thomas LLC	465.34	301
PROD16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	256
PROD17	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy and Wilson	144.53	367
PROD18	Women's Denim Jeans	Clothing	Bottoms	Guerra-Hendrix	245.78	343
PROD19	Knee Support Brace	Sports	Equipment	May-Reese	21.34	65
PROD20	Outdoor Sports Water Bottle	Sports	Accessories	Soto-Ellis	131.51	264
PROD21	Multi-Purpose Storage Box	Home	Storage	Gardner-Montes	123.46	76
PROD22	Carbon Fiber Tennis Racket	Sports	Equipment	Flores, Pineda and Rodrigue	484.62	432
PROD23	Men's Formal Shirt	Clothing	Tops	Moon, Mora and Santos	248.82	360
PROD24	Foldable Camping Chair	Sports	Outdoor	Cook Ltd	166.06	340
PROD25	Women's Scarf Set	Clothing	Accessories	Brown and Sons	174.02	268
PROD26	Wireless Bluetooth Speaker	Electronics	Audio	Reed-Schmidt	322.41	213
PROD27	Decorative Floor Vase	Home	Decor	Khan, Jones and Paul	250.02	378
PROD28	Sports Stopwatch Timer	Sports	Accessories	Burns-Hansen	38.76	137
PROD29	Fabric Sofa Throw Blanket	Home	Decor	Hoffman Group	218.48	276
PROD30	Stainless Steel Cutlery Set	Home	Kitchen	Stevens, Fernandez and Ma	19.52	57
PROD31	Basketball	Sports	Team Sports	Nolan-Snow	204.39	281
PROD32	Vitamin C Face Serum	Beauty	Skincare	Griffin Ltd	283.83	244
PROD33	Modern Wall Clock	Home	Decor	Weiss-Watson	179.77	305
PROD34	Gaming Mechanical Keyboard	Electronics	Computer	Lucero LLC	419.28	86
PROD35	Bedroom Table Lamp	Home	Lighting	Jones LLC	382.22	329
PROD36	Women's Summer Dress	Clothing	Tops	Smith-Mills	223.43	154

Find, Remove Duplicates, Handle Missing, Inconsistent Data, Correct Formatting errors, and Validate Data

REMOVE DUPLICATE DATA: Data-->Data Tool-->Remove Duplicates

COST AND STOCK COLUMN HANDLE MISSING DATA:
COST: IF(ISBLANK(F2),AVERAGE(F2:F103),F2)
STOCK:IF(ISBLANK(G2),AVERAGE(G2:G103),G2)

CORRECT FORMATTING: PROPER(TRIM(CLEAN(B2)
PROPER(TRIM(CLEAN(C2) PROPER(TRIM(CLEAN(D2)
PROPER(TRIM(CLEAN(E2)

Customer_Dim | Product_Dim | Store_Dim | Sales_Fact | DESCRIPTIVE STATISTICS | PIVOT TABLE | DASHBOARD | xloo

STORE DIM

Store_ID	Store_Name	Region	City	Store_Type
STORE0	Rivera-Warner	East	Washingtonfurt	Flagship
STORE1	Martinez, Adkins and Johnson	North	Danielmouth	Flagship
STORE2	Black-Horn	South	North Nancyview	Outlet
STORE3	Johnson-Wright	North	Hoodberg	Outlet
STORE4	Alexander Group	South	Port Sharon	Online
STORE5	Gray Ltd	North	New Michaelchester	Outlet
STORE6	Benson-Elliott	East	Port Josephberg	Online
STORE7	Martinez-Mullins	South	Williamsfurt	Online
STORE8	Bean, Jones and Park	West	Thompsonborough	Online
STORE9	Moore, Padilla and Butler	South	East Johnstad	Flagship
STORE10	Hill, Wagner and Ballard	East	East Davidville	Online
STORE11	Aguirre-Keller	South	Lake Michael	Online
STORE12	Thomas, Bridges and Harris	North	Sabrinatown	Online
STORE13	Gardner, Humphrey and Contrera	East	Johnsonville	Flagship
STORE14	Mcdonald Inc	South	Christopherland	Outlet
STORE15	Shelton PLC	North	East Jenniferstad	Flagship
STORE16	Frederick-Wright	West	New Jimmymouth	Flagship
STORE17	Holloway Ltd	North	Bensonhaven	Flagship
STORE18	Guerrero-Schmitt	East	East Stevechester	Flagship
STORE19	Mann-Austin	East	Vasquezborough	Online

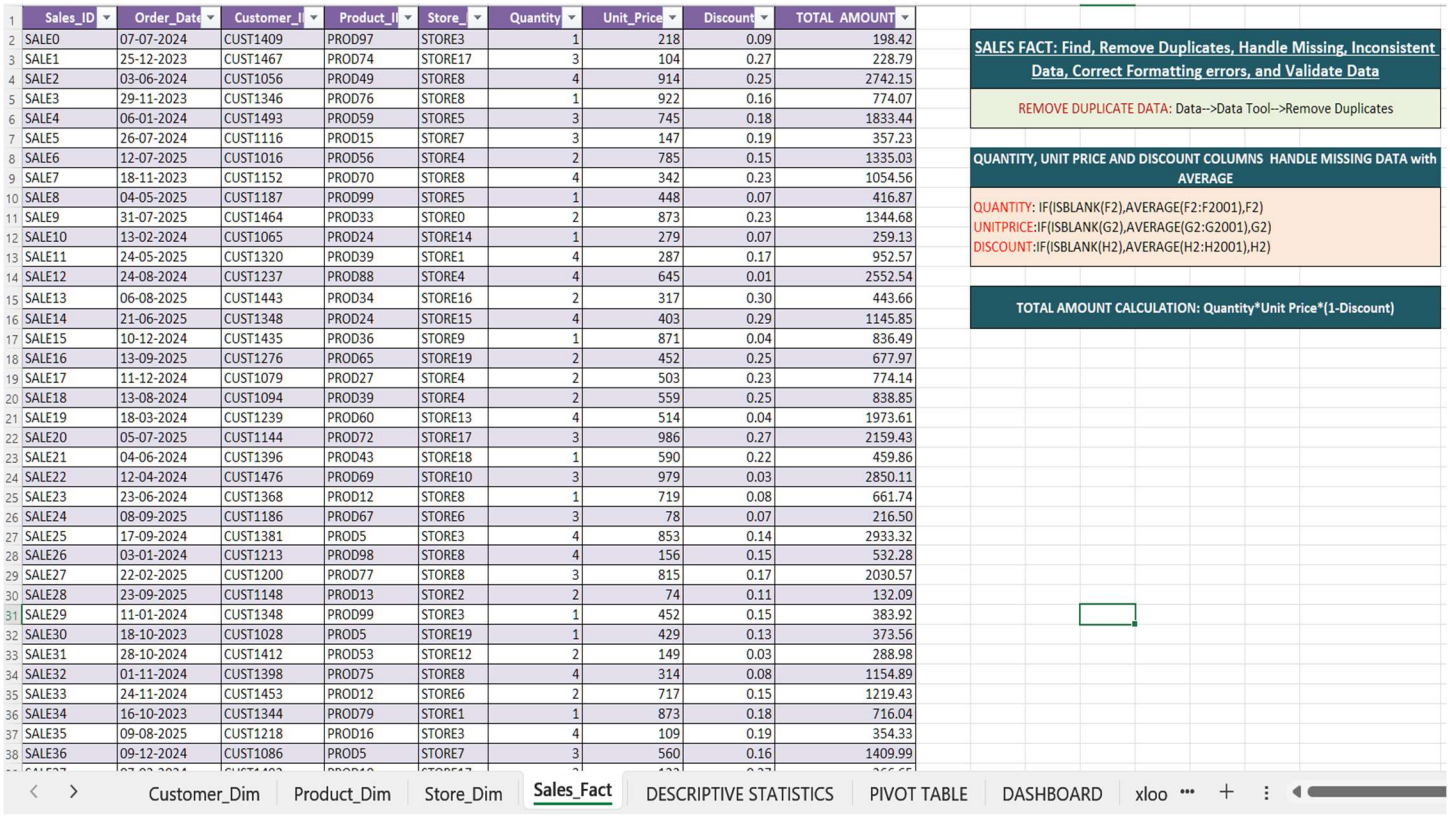
STORE DIM: Find, Remove Duplicates, Handle Missing, Inconsistent Data, Correct Formatting errors, and Validate Data

MOVE DUPLICATE DATA: Data-->Data Tool-->Remove Duplic

CORRECT FORMATTING: PROPER(TRIM(CLEAN(B2)
PROPER(TRIM(CLEAN(D2)

Customer_Dim | Product_Dim | Store_Dim | Sales_Fact | DESCRIPTIVE STATISTICS | PIVOT TABLE | DASHBOARD | xloo

SALES FACT



Sales_ID	Order_Date	Customer_I	Product_I	Store_	Quantity	Unit_Price	Discount	TOTAL AMOUNT
SALE0	07-07-2024	CUST1409	PROD97	STORE3	1	218	0.09	198.42
SALE1	25-12-2023	CUST1467	PROD74	STORE17	3	104	0.27	228.79
SALE2	03-06-2024	CUST1056	PROD49	STORE8	4	914	0.25	2742.15
SALE3	29-11-2023	CUST1346	PROD76	STORE8	1	922	0.16	774.07
SALE4	06-01-2024	CUST1493	PROD59	STORE5	3	745	0.18	1833.44
SALE5	26-07-2024	CUST1116	PROD15	STORE7	3	147	0.19	357.23
SALE6	12-07-2025	CUST1016	PROD56	STORE4	2	785	0.15	1335.03
SALE7	18-11-2023	CUST1152	PROD70	STORE8	4	342	0.23	1054.56
SALE8	04-05-2025	CUST1187	PROD99	STORE5	1	448	0.07	416.87
SALE9	31-07-2025	CUST1464	PROD33	STORE0	2	873	0.23	1344.68
SALE10	13-02-2024	CUST1065	PROD24	STORE14	1	279	0.07	259.13
SALE11	24-05-2025	CUST1320	PROD39	STORE1	4	287	0.17	952.57
SALE12	24-08-2024	CUST1237	PROD88	STORE4	4	645	0.01	2552.54
SALE13	06-08-2025	CUST1443	PROD34	STORE16	2	317	0.30	443.66
SALE14	21-06-2025	CUST1348	PROD24	STORE15	4	403	0.29	1145.85
SALE15	10-12-2024	CUST1435	PROD36	STORE9	1	871	0.04	836.49
SALE16	13-09-2025	CUST1276	PROD65	STORE19	2	452	0.25	677.97
SALE17	11-12-2024	CUST1079	PROD27	STORE4	2	503	0.23	774.14
SALE18	13-08-2024	CUST1094	PROD39	STORE4	2	559	0.25	838.85
SALE19	18-03-2024	CUST1239	PROD60	STORE13	4	514	0.04	1973.61
SALE20	05-07-2025	CUST1144	PROD72	STORE17	3	986	0.27	2159.43
SALE21	04-06-2024	CUST1396	PROD43	STORE18	1	590	0.22	459.86
SALE22	12-04-2024	CUST1476	PROD69	STORE10	3	979	0.03	2850.11
SALE23	23-06-2024	CUST1368	PROD12	STORE8	1	719	0.08	661.74
SALE24	08-09-2025	CUST1186	PROD67	STORE6	3	78	0.07	216.50
SALE25	17-09-2024	CUST1381	PROD5	STORE3	4	853	0.14	2933.32
SALE26	03-01-2024	CUST1213	PROD98	STORE8	4	156	0.15	532.28
SALE27	22-02-2025	CUST1200	PROD77	STORE8	3	815	0.17	2030.57
SALE28	23-09-2025	CUST1148	PROD13	STORE2	2	74	0.11	132.09
SALE29	11-01-2024	CUST1348	PROD99	STORE3	1	452	0.15	383.92
SALE30	18-10-2023	CUST1028	PROD5	STORE19	1	429	0.13	373.56
SALE31	28-10-2024	CUST1412	PROD53	STORE12	2	149	0.03	288.98
SALE32	01-11-2024	CUST1398	PROD75	STORE8	4	314	0.08	1154.89
SALE33	24-11-2024	CUST1453	PROD12	STORE6	2	717	0.15	1219.43
SALE34	16-10-2023	CUST1344	PROD79	STORE1	1	873	0.18	716.04
SALE35	09-08-2025	CUST1218	PROD16	STORE3	4	109	0.19	354.33
SALE36	09-12-2024	CUST1086	PROD5	STORE7	3	560	0.16	1409.99

SALES FACT: Find, Remove Duplicates, Handle Missing, Inconsistent Data, Correct Formatting errors, and Validate Data

REMOVE DUPLICATE DATA: Data-->Data Tool-->Remove Duplicates

QUANTITY, UNIT PRICE AND DISCOUNT COLUMNS HANDLE MISSING DATA with AVERAGE

QUANTITY: IF(ISBLANK(F2),AVERAGE(F2:F2001),F2)
UNITPRICE:IF(ISBLANK(G2),AVERAGE(G2:G2001),G2)
DISCOUNT:IF(ISBLANK(H2),AVERAGE(H2:H2001),H2)

TOTAL AMOUNT CALCULATION: Quantity*Unit Price*(1-Discount)

Customer_Dim | Product_Dim | Store_Dim | Sales_Fact | DESCRIPTIVE STATISTICS | PIVOT TABLE | DASHBOARD | xloo

Visualization and Insights Arrived with screenshot of visualization Report/Dashboard screenshot:

- **Descriptive analysis:**

  This explains that across 2,000 sales transactions in the dataset, the total revenue reached ₹21,70,101. It shows that Sports and Electronics categories generated the highest sales amounts, and female customers (57%) made more purchases compared to male customers (43%).

- **Diagnostic Analysis:**

  This helps uncover the reasons behind why sales were higher or lower in specific stores and regions. For instance, it analyzes how online and flagship store types drove higher quantities and revenues, whereas outlet stores experienced lower sales performance.

- **Predictive Analysis:**

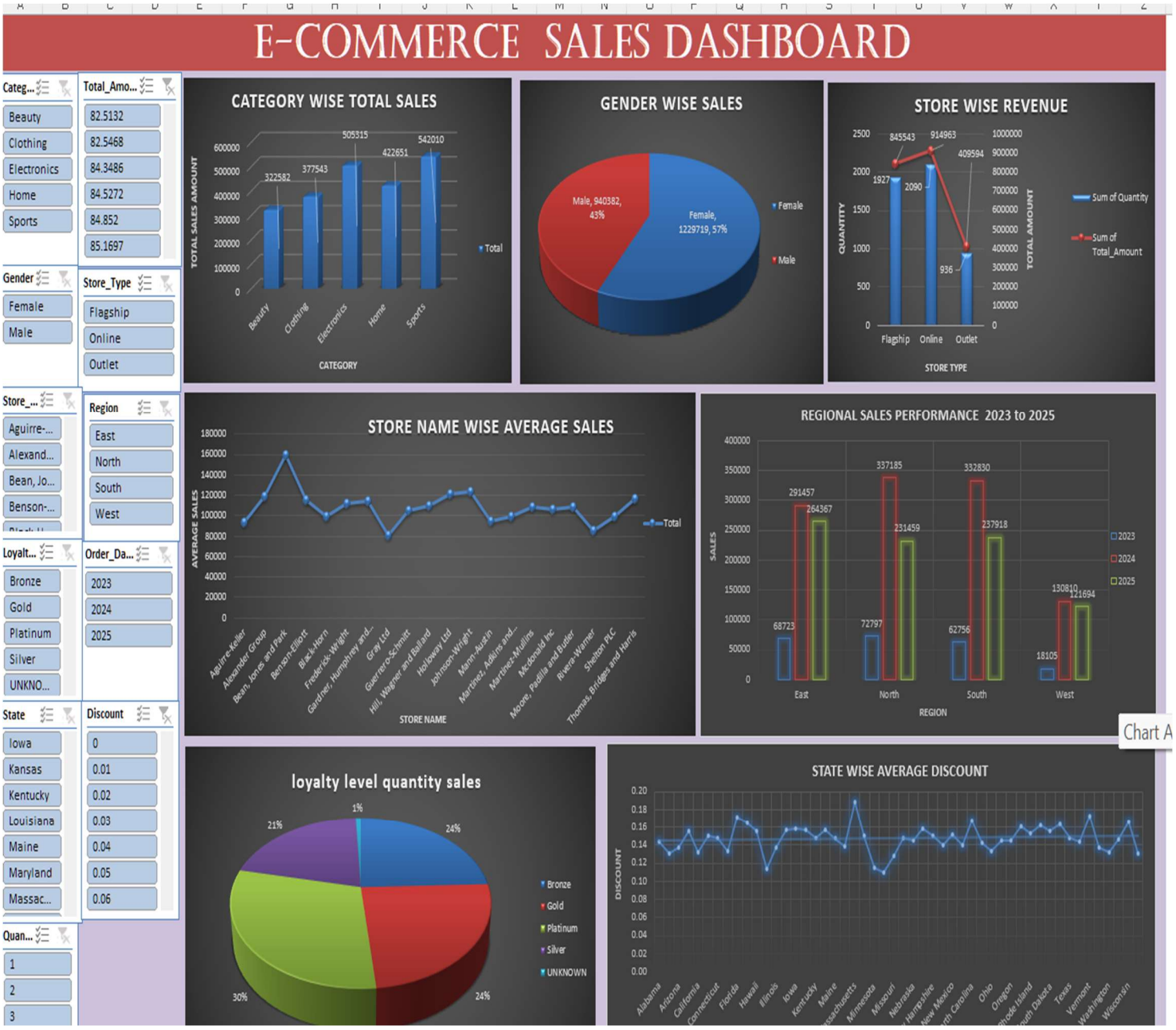
  This forecasts future sales trends based on historical sales data from 2023 to 2025. By observing how sales peaked in 2024 and shifted slightly across certain regions in 2025, it helps anticipate customer purchasing power and demand for the upcoming period.

- **Prescriptive Analysis:**

  This recommends actions to improve the business. Since Platinum and Gold loyalty customers deliver higher value to the business, it guides that offering targeted discounts and benifits to them can further increase future sales.

DASHBOARD IMAGE:

E-COMMERCE SALES DASHBOARD



FINAL 4 TYPE OF INSIGHTS AS STORYTELLING:

Descriptive Insight Story:

Our store areas are always busy with activity, particularly Sports and Electronics items have emerged as customer's top choices, crossing over ₹500,000 in sales and driving up overall business revenue.

Diagnostic Insight Story:

Customer behavior demonstrates that store location and channel type matter significantly. Because online and flagship stores are easily accessible, customers prefer them, whereas outlet stores lag behind due to weaker promotional strategies.

Predictive Insight Story:

A close look at past sales charts indicates that the massive sales surge seen in mid-2024 may experience minor fluctuations in certain regions by 2025, yet the East and North regions will continue to maintain healthy business growth moving forward.

Prescriptive Insight Story:

Our true strength lies within our loyal customer base. By distributing exclusive discount coupons to Platinum and Gold members and applying proper pricing/discount strategies in low-performing stores, we can multiply our overall revenue.

CONCLUSION:

This e-commerce capstone project successfully accomplished everything from data cleaning and transformation to building a Star Schema data modeling structure. The precise insights derived through Excel and pivot tables clearly highlight the areas requiring focus. By leveraging these analytical findings to refine marketing strategies and enhance customer service, businesses can achieve exceptional commercial growth in the future.